



Assignment Overview: Data Governance

Estimated time: 4 minutes

Learning objectives

- Draft a comprehensive data governance plan to address compliance, data protection, and regulatory considerations
- Identify key data governance roles, responsibilities, and policies, to ensure proper data management and legal compliance.

Introduction

- In phase 1, you comprehensively analyzed **TrendyThreads'** and **FashionMart's** current data architectures. This helped you identify key inefficiencies, integration challenges, and areas where best practices could be applied to improve the overall architecture.
- In phase 2, you developed a consolidated data architecture for the merged entity. This included creating entity-relationship (ER) diagrams, defining table structures, and optimizing the design to remove performance bottlenecks and streamline data flow across systems.
- Phase 3 focused on executing the data migration strategy. You facilitated the data transition between RDBMS and NoSQL platforms, ensuring a phased migration approach that maintained data integrity and minimized disruptions.
- In phase 4, the focus shifted to seamless data integration. You implemented ETL/ELT scripts, built robust data pipelines, and designed a scalable data warehouse schema to support real-time analytics and reporting. Additionally, specialized pipelines were created to support fraud detection, ensuring secure and timely data flow.

Phase 5: Draft data governance plan for FutureMart

Phase 5 will focus on creating a comprehensive data governance plan for **FutureMart**. This plan will ensure proper data management, security, and compliance by:

- Establishing clear governance objectives aligned with business goals
- Defining a governance framework and assigning roles and responsibilities
- Creating policies and processes for data classification, access control, and ongoing quality management

Tasks

Section 1: Define the data governance purpose and scope for FutureMart

In this section, you'll define the foundational aspects of **FutureMart's** data governance plan, focusing on its purpose and scope to ensure integrity, privacy, and effective data use.

- **Purpose of data governance:** Articulate the main objective of implementing data governance at **FutureMart**
- **Scope of data governance:** Identify business areas covered under data governance, including sales, marketing, customer service, inventory, and finance

Section 2: Draft data governance objectives for FutureMart

In this section, you'll outline key objectives to ensure that **FutureMart** maintains data accuracy, security, and accessibility aligned with business goals.

- **Data quality objectives:** Ensure that data is complete, up-to-date, and accurate
- **Data security objectives:** Protect personal and sensitive information from unauthorized access
- **Data compliance objectives:** Align with legal and regulatory requirements
- **Data availability objectives:** Ensure that data is accessible to authorized users
- **Data consistency objectives:** Maintain consistency across systems and platforms
- **Data usage objectives:** Enable data-driven decision-making across departments

Section 3: Create the data governance framework (Optional)

This section is optional and for practice only. You'll build a data governance framework including roles, responsibilities, and core policies.

List and describe the responsibilities of:

- Data Governance Steering Committee
- Data stewards
- Data owners
- Data users (e.g., Data Analysts)

Draft the following governance policies:

- **Data collection policy:** Methods for collecting/storing customer data
- **Data usage policy:** Guidelines for marketing and sales data use
- **Data retention policy:** Define retention duration and archival/deletion processes
- **Data privacy and security policy:** Include data encryption, masking, and access control

Section 4: Establish data management practices

In this section, you'll implement robust data management practices to maintain integrity, security, and accessibility across the organization.

- **Data classification practices:** Categorize data by sensitivity (public, internal, confidential, restricted)
- **Data quality management practices:** Define validation processes to ensure accuracy and completeness
- **Data integration practices:** Establish standards for integrating data from multiple sources
- **Data documentation practices:** Ensure thorough documentation for consistent data structure and accuracy

Section 5: Define data security and privacy policies

In this section, you'll create policies to protect **FutureMart's** data during storage, access, and incidents.

- **Data encryption and masking policy:** Secure data in storage and during transmission

- **Data access control policy:** Restrict access to sensitive data to only authorized users
- **Data incident management policy:** Define protocols for handling breaches and unauthorized access events

Section 6: Define compliance and legal considerations policies

In this section, you'll create policies to ensure that **FutureMart** remains legally compliant and accountable.

- **Regulatory compliance policies:** List applicable laws and regulations for **FutureMart's** data usage
- **Data auditing and reporting policy:** Establish procedures for regular audits and compliance checks
- **Data retention and disposal policy:** Set data retention periods and secure disposal protocols based on legal and business needs

Section 7: Draft additional components of the data governance plan (Optional)

This optional section is for practice and won't be graded. You can enrich the governance plan by covering implementation, tools, and performance.

- **Data governance tools and technologies:** Identify platforms/software for data management and monitoring
- **Data governance training and awareness:** Propose training programs and awareness initiatives
- **Performance metrics and reporting:** Develop data quality, security, and compliance KPIs
- **Phased implementation plan:** Allocate resources (tools, personnel, budget) and outline continuous improvement processes

Deliverables

By the end of **Module 5**, you will have completed the following deliverable:

- The data governance plan for FutureMart.